

FACULTY OF INFORMATION TECHNOLOGY

BBT 2104 – INTEGRAL CALCULUS

TIME: 2 Hours

DATE: Thursday, 9th February 2017

Instruction

Answer QUESTION ONE and any other TWO questions

QUESTION ONE (30 Marks)

- a) Evaluate the given integral $\int x^4(2x^5 + 5)^6 dx$ (3 marks)
- b) Determine the area between the curve $y = x(x - 3)$ and the $x - axis$ from $x = 0$ to $x = 3$. (4 marks)
- c) Use an appropriate method to evaluate
 - i. $\int \sin^3 x \cos^2 x dx$ (5 marks)
 - ii. $\int_0^{\frac{\pi}{4}} (1 + \cos 4x)^{0.5} dx$ (4 marks)
- d) Find the general solution of the differential equation $x \frac{dy}{dx} + 2 \frac{dy}{dx} = y + 1$ (4 marks)
- e) Determine the volume of revolution formed by rotating the area enclosed by the curve $y = x + x^2$, the $x - axis$ and the points $x = 2$ and $x = 3$ through one revolution about the $x - axis$. (3 marks)
- f) Use partial fractions to evaluate $\int \frac{5x-3}{(x+1)(x-3)} dx$ (4 marks)
- g) Use Taylor's theorem to obtain a series expansion for $\cos(x + \frac{\pi}{3})$ up to and including terms in x^3 (3 marks)

QUESTION TWO (20 Marks)

- a) Find the area between the curve $y = x(4 - x)$ and the $x axis$ from $x = 0$ to $x = 5$. [5 Marks]
- b) Find the area enclosed between the curve $y^2 = 9 - x$ and the $y axis$ [5 Marks]
- c) Evaluate the following integrals :
 - i. $\int \sin^2 x \cos^5 x dx$ [5 Marks]

II. Use integration by parts to evaluate $\int_1^2 x^2 \ln x \, dx$ [5 Marks]

QUESTION THREE (20 Marks)

a) Evaluate using the method of change of variable [5 marks]

$$\int x^2(2x^3 + 3)^5 dx$$

b) Find the solution of the differential equation $xy' + 2y' - 1 = y$ giving your answer in terms of y . [5 Marks]

c) Use Taylor's theorem to obtain a series expansion for $\cos(x + \frac{\pi}{3})$ up to and including terms in x^3 . [10 Marks]

QUESTION FOUR (20 Marks)

a) Find the approximate value for ;

i. $\int_0^1 e^x dx$ using the trapezium rule with 5 strips [10 Marks]

ii. Given $y = f(x)$ and that for given values of x , the corresponding values for y are shown in the table below, use Simpson's rule with 8 strips to approximate

$$\int_{-1}^7 f(x) dx$$

x	-1	0	1	2	3	4	5	6	7
y	17	10	5	2	1	2	5	10	17

[10 Marks]

QUESTION FIVE (20 Marks)

a) Find the volume of the solid generated by rotating about the x axis the region bounded by $y = x^2$, the x axis and the lines $x = 1$ and $x = 2$ [3 Marks]

b) Suppose we have a cone of base radius r and vertical height h , imagine the cone being formed by rotating a straight line through the origin by an angle of 360 degrees about the x axis. Show

that the volume of the cone is given by $\frac{\pi r^2 h}{3}$ [7 Marks]

c) Solve the following differential equations

i. $3(y^2 + 1)dx + 2xy \, dy = 0$ [5 Marks]

ii. $y' = \frac{1}{x^2 y}$ [5 Marks]